

CULTURE AND CONNECTIVITY INTERTWINED: VISUALIZING ORGANIZATIONAL FIELDS AS RELATIONAL STRUCTURES *AND* MEANING SYSTEMS ☆

Achim Oberg, Valeska P. Korff and Walter W. Powell

ABSTRACT

Organizational fields are shaped by both the relations that organizations forge and the language they express. The structure and discourse of organizational fields have been studied before, but seldom in combination. We offer a methodological approach that integrates relations and expressions into a comprehensive visualization.

By mapping networks and discourse as co-constitutive, the method illuminates the mechanisms active in organizational fields. We utilize social impact evaluation as an issue field shaped by the presence of an interstitial community, and compare this structure with simulated alternative field configurations.

The simulations reveal that variation in organizations' openness to adopting concepts from adjacent meaning systems alters field configurations: differentiation

☆ Note: Full color versions of the figures included in this chapter can be found in the online version of this volume.

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manifests under conditions of low overall openness, whereas moderate receptivity produces hybridizations of discourses and sometimes the emergence of an interstitial community that bridges domains. If certain organizations are open while others remain focused on their original discourse, then we observe integration in the discursive domain of the invariant organizations.

The observations from the simulations are represented by visualizing organizational fields as topographies of meaning, onto which interorganizational relations are layered. This representation localizes organizations and their interactions in a cultural space while emphasizing how meanings of relationships and organizational expressions vary with different field configurations. By adding meaning to network data, the resulting maps open new perspectives for institutional research on the adaptation, translation, and diffusion of concepts.

Keywords: Organizational fields; networks; discourse; simulations; interstitial organizations; visualization

INTRODUCTION

Organizational fields are often described as communities of organizations that share a common meaning system, and whose participants interact frequently and consequentially with one another (DiMaggio & Powell, 1983; Scott, 1994). Many empirical studies focus on interactions among organizations as the process by which members of a field exchange resources and ideas. However, the spread and elaboration of concepts among members of a field are not only channeled through direct contacts, but also occur through the evolution of cultural categories and codes expressed in a common discourse (Lawrence & Phillips, 2004; Pachucki & Breiger, 2010). Approaches differ with regard to their emphasis on either relationships or language as constitutive elements, reflecting the variety of analytical lenses through which fields have been examined and theorized about. Despite the focus on different elements, the perspectives are in principle compatible, as Zietsma, Groenewegen, Logue, and Hinjns (2017) argue in a recent review.

To foster such integration, we introduce an analytical approach that combines network and discursive perspectives. To examine organizational fields as co-constitutive systems of relations and meaning, we apply data visualization tools that render complex, multidimensional data accessible and interpretable. In the natural sciences, visualization has long been appreciated as a central tool to aid discovery, understanding, and communication. These qualities are increasingly being valued in sociological analyses as well (Moody & Healy, 2014). We create maps by successively layering different types of data – particularly relational and discursive data, but also information on demographic properties of

organizations — to produce a multilevel representation of a specific organizational field's configuration. The involved organizations are identifiable as units, both in terms of their connections to others and as occupiers of distinctive cultural positions. By revealing a field's relational and discursive structure, our representation enables the systematic comparison of patterns of structuration and the identification of different types of field configurations.

In a previous work, we applied this approach to analyze the debate over social impact in the nonprofit sector. We positioned 369 organizations on a triangular plane depending on the extent to which they drew on scientific, managerial, or associational concepts in their online self-representations. Hyperlinks were then layered on this discursive topography to add a relational dimension, producing a map as shown in Fig. 1 (Korff, Oberg, & Powell, 2015). The observed issue field is characterized by the presence of organizations at the interface between the domains of science, management, and civil society. These interstitial organizations share a common language and are closely connected, forming a community with dense network ties that convey ideas across spheres (Korff, Oberg, & Powell, 2017). Such a configuration is distinctive: fields anchored by interstitial communities combine diverse discourses and ties formed across boundaries. This structure can also exert substantial influence. The debate on social impact, although heterodox, has attained almost compulsory features, drawing in a variety of organizations, including associations and foundations, consulting firms, think tanks, international organizations, and local social service providers. These diverse organizations adopt a composite language derived from multiple meaning systems, and connect to others that share the language, even if they differ in organizational form.

The empirical application of our visualization approach revealed a pattern in which network centrality and influence on discourse are deeply intertwined, reflecting a situation that can be understood as characteristic of organizational interactions in the digital era (Castells, 2000; Powell, Oberg, Korff, Oelberger, & Kloos, 2017). At the same time, however, these patterns are by no means universal. Organizations are the locus where many of the critical activities of modern society occur. Organizational processes generate a variety of divergent outcomes, including competition, collaboration, innovation, and control. Alternative field configurations can therefore be conceived, raising the question of what outcomes are likely to manifest under different conditions. More specifically, if openness to adopt other concepts is central to the emergence of interstitial communities, then what processes underlie the formation of other types of field configurations? To address this question, we contrast the observed empirical pattern of an interstitial community with alternative configurations that draw upon several well-understood social processes.

Three such mechanisms and corresponding field configurations are presented in detail: 1. *Differentiation*, a state of segregation in which organizations of a particular form are concentrated in certain discursive domains and connected predominantly with those that are similar; 2. *Recombination*, a situation

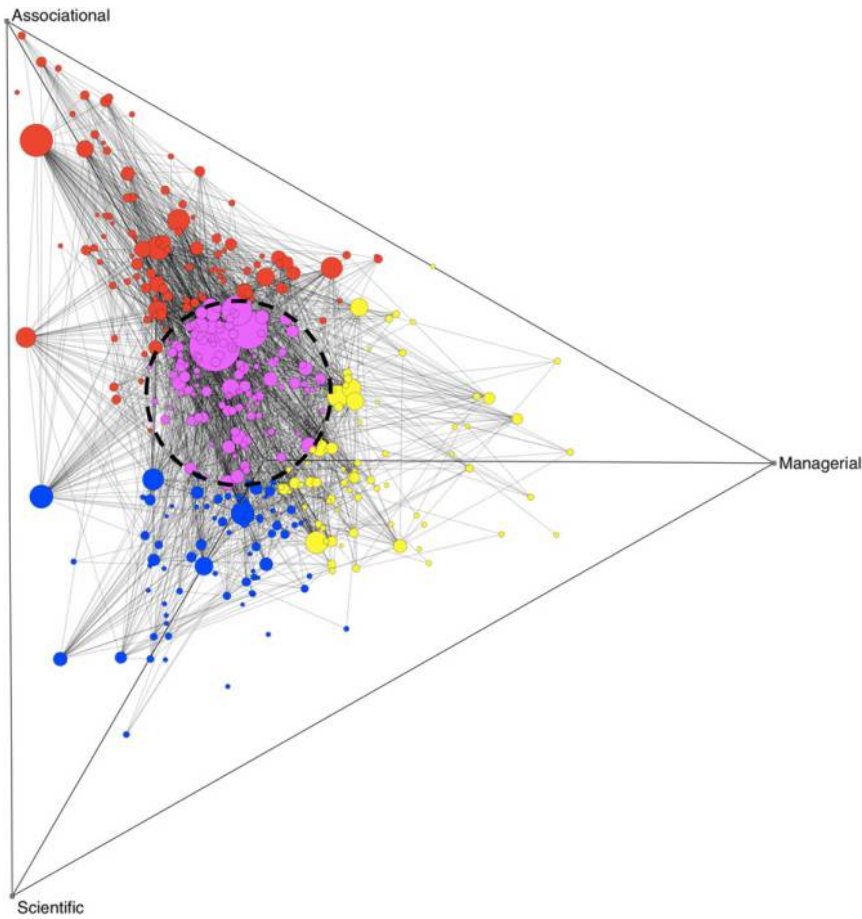


Fig. 1. Relational and Discursive Dimensions of the Social Impact Debate in the United States.

characterized by porous boundaries that occur when organizations engage in other discourses and create new cultural content; and 3. *Integration*, a position characterized by the authority of one orientation, which pulls in organizations of various forms, irrespective of their original orientation.

Representing such dynamics as stylized mechanisms has become a model for theorizing that emphasizes verbs more than nouns (Hedström & Bearman, 2009; Padgett & Powell, 2012). From a methodological viewpoint, these prototypical configurations illustrate how different relational and cultural properties produce distinctive maps, underscoring how visualization can serve as an analytical tool. Rather than explore diverse empirical settings and deal with

associated challenges in terms of selecting suitable cases and obtaining usable data, we apply data simulations to reflect different social processes and create corresponding maps. As a method, simulations allow us to simultaneously illustrate our methodological approach and examine how different mechanisms produce variation in the configuration of hypothesized fields.

We begin with a default field in which 300 organizations representing three forms are located in a triangular plane, suspended between three pillars representing different discourses and their meaning systems. We then vary the degree to which the field members are open to adopt elements from other meaning systems into their self-representations, and hence disposed to shifting their position within the topography of meaning. Our empirical map of an interstitial community, derived from webpages in 2011, serves as a basis for comparison and contrast.

Generating data through simulations has several advantages. A significant obstacle to combining network and discourse analysis is the difficulty of collecting and analyzing very different yet equally hard-to-obtain data on content and connections in one project. Simulations can overcome this challenge because they capture relational features and account for discursive orientations. They further enable the controlled variation of these dimensions, thus allowing for the systematic exploration of the effects of and interactions between networks and discourse (Schelling, 1969). As a result, the application of simulations can show how relational structure and discourse are intertwined and produce diverse patterns through their interplay.

Our mapping technique makes these co-constitutive systems visible. After elaborating on different conceptual field configurations, we outline the necessary steps to simulate and visualize them. We then present the simulated scenarios and highlight the specific connections between meaning systems and social relationships. The ensuing discussion focuses on how such simulations can be used as a backdrop for analyzing real-world data, and how visualizations may serve to reveal how organizational features, relationships, and discourses are intertwined. We conclude by discussing how the combination of meaning and networks could revive research on the diffusion of ideas by quantifying and testing ideas about editing and translation.

TRADITIONS IN THE ANALYSIS OF FIELDS: NETWORK AND DISCOURSE ANALYSIS

Organizational fields as objects of empirical inquiry have traditionally been studied either as webs of relations or as meaning systems. Researchers emphasizing the relational dimension of organizational fields have the rich repertoire of network analysis methods and techniques at their disposal (De Nooy, Mrvar, & Batagelj, 2012), which can illustrate and analyze relations and

network structures, even in large interconnected fields (Bail, 2014; Foster, Rzhetsky, & Evans, 2015; Powell, White, Koput, & Owen-Smith, 2005). The advantage of this approach is that both local effects and the entire structure of a network can be analyzed simultaneously (Scott, 2017; Wasserman & Faust, 1994).

Network analysis offers an in-depth picture of a crucial dimension of organizational fields: the relational structure formed by and the underlying interactions among organizations. Analyses of complex relations typically require the characteristics of specific organizations to be condensed into relatively simple categorizations. Network effects become observable when organizations are treated as nodes; on the other hand, factors related to an organization's discursive orientation recede in importance. Such omissions run the risk of overestimating the importance of ties between organizations while neglecting effects derived from shared language or ideology. The analysis of relational structures provides just one perspective on the multifaceted phenomenon of how fields are structured.

Researchers interested in meaning systems and the diffusion of norms within organizational fields typically place greater emphasis on analyzing language (Hoffman, 1999). Industry reports, trade newspapers or magazines, and documents issued by field members directly, such as press releases and organizational reports, provide rich sources of text data (Phillips & Oswick, 2012; White, 2004). Discursive analyses of language, whether interpretive or quantitative, represent efforts at reconstructing a field's meaning system based on available text and visual materials (Grimmer & Stewart, 2013; Mohr, 1998; Mohr & Ghaziani, 2014).

Discourse analysis facilitates an understanding of a field's language by offering insights into the codes and meaning system that underlie and shape interactions. Although methods exist that preserve the producers of text – topic modeling is one example (Mohr & Bogdanov, 2013) – the focus on texts and other artifacts typically downplays the role of the creators, i.e., the organizations that directly contribute to the discourse. Usually, only certain types of documents issued by particular members of a field are included in the analysis (see Suddaby & Greenwood, 2005). The exclusive focus on industry magazines or business reports discounts organizations such as regulatory bodies, consultants, critics, audiences, academic experts, and funders that are also active participants in and shapers of a field's discourse. Generally, discourse analysis focuses on examining units of text to identify central and peripheral concepts and to understand constructions of meaning (Mohr, 1994). This elevation of the analysis to the level of the discourse as a system shifts the focus toward the cultural content and the relations between associated codes and symbols. Concepts, their classifications, and the boundaries between categories take center stage, whereas the producers of texts and their features and relations recede into the analytical background.

Toward a Combination of Perspectives

Despite a common interest in the configuration of fields, network and discourse analysis place different emphasis on either relationships or language and symbols as constitutive elements uniting a community of organizations. Their empirical methods reflect and further solidify this divide. Structuralists focus on the relational configurations of organizations as closely interconnected nodes, while assuming but not measuring cultural consensus (Davis & Greve, 1997; Owen-Smith & Powell, 2004; Vedres & Stark, 2010). In contrast, interpretivist scholars draw on discursive approaches that identify topographies of meaning but leave networks of affiliation in the background (Meyer & Höllerer, 2010; Vaara & Monin, 2010). As a result, culture and networks have been treated as separate areas of inquiry.

This taken-for-granted separation has been called into question (Bail, 2012; Kirchner & Mohr, 2010). McLean (2007, p. 6) begins his absorbing *Art of the Network* with the question, “Where is the culture in networks?” Others also argue that network analysis has been largely blind to the social content of the relations that are so carefully mapped (Emirbayer, 1997; Emirbayer & Goodwin, 1994; Fligstein, 2013). One exception is studies that disentangle the effects of selection versus influence on the structure of social networks (Aral, Muchnik, & Sundararajan, 2009; Steglich, Snijders, & Pearson, 2010). Underlying these efforts is the question of the extent to which commonalities in behavior or orientation among closely connected nodes are derived from a proclivity to form ties with similar others (McPherson, Smith-Loving, & Cook, 2001) or from a process of contagion in which traits, opinions, and practices are transmitted through social ties (Friedkin, 2001). However, the idea that cultural codes and concepts may diffuse through the evolution of a common language and meaning system – even in the absence of direct ties – also remains in the background in these formulations.

An effort to integrate network and discourse perspectives of organizational fields should incorporate both structure and language as field-configuring dimensions. Mapping field members and their relations within a topography of meaning allows for locating organizational ties and interactions in cultural space. Our approach hinges on the idea that organizational fields do not exhibit one singular framework without variation but instead comprise and connect multiple meaning systems. The issue field that encompasses social impact analysis, for example, brings together a managerial frame that alludes to performance and outputs; a scientific approach with an emphasis on measurement and data; and civic values such as empowerment, justice, and social change. Each frame or meaning system provides concepts and practices that participants appropriate in crafting their approaches and orientations. The extent to which an organization borrows from each of the respective domains determines its position on the topography, revealing its location both vis-à-vis the three

meaning systems as well as relative to other field members. In our empirical case, the majority of organizations draw from all three frames, combining managerial terms with scientific and civic concepts on their webpages (Korff et al., 2015). Located at the intersection between meaning systems, these organizations can be understood as speaking the same composite language, being familiar with the same concepts, and perhaps championing similar ideas. When relational properties – the ties between organizations – are layered upon this topography, the structural and discursive configuration of the field becomes visible. In the example above, this approach revealed a densely connected community at the interstice that serves as a conversational bridge between domains. As Furnari (2014) argues for micro-level interactions across field boundaries, a central factor shaping an interstitial space is openness to adopting elements of multiple meaning systems among the involved participants. However, such openness to shifting one's position in the topography of meaning and establishing ties to organizations of different forms is not a universal feature but one that may vary decidedly between organizational fields. As we argue below, different social processes are likely to facilitate the manifestation of alternative field configurations.

SOCIAL PROCESSES OF FIELD CONFIGURATION

A field is a social setting in which the position of entities reflects the interactions between specific rules and properties of its members (Bourdieu & Wacquant, 1992). Accordingly, organizations and their behaviors and identities are both shaped by and influence the field they inhabit. Based on the assumption that organizations within a field strive for legitimacy, DiMaggio and Powell (1983) identified three mechanisms – mimetic, coercive, and normative isomorphism – that characterize interactions among field participants. This view of organizations as both subject to, and producers of, field dynamics proved appealing and sparked interest in the study of mechanisms that characterize interorganizational relations.

We follow this tradition with our simulations, suggesting three possible social processes that influence both the relational and discursive aspects of fields. *Differentiation* is a process by which individual organizations, or a select group, carve out unique identities, distinct from others. *Recombination* is a form of borrowing, or import-export relations, as organizations freely draw on specific aspects of other organizations and assemble newer practices of others with their extant ones. *Integration* entails the whole-cloth adoption of templates or packages, which is a process that induces heightened similarity among participants.

Differentiation produces diversity and segregation within a field. Phillips and Zuckermann (2001) offer a canonical account of this process in their analysis of

middle status conformity – the long-held assumption that conformity is high at the middle and low at either end of a status order. Drawing on empirical studies of the Silicon Valley legal services field and securities analysts in the investment-banking field, the authors find a legitimacy-based process of differentiation. The mechanisms involved are quite clear: conformity with respect to certain norms but not others (a) increases with an actor's desire to be recognized as a member of a social category, and (b) decreases with an actor's sense of security as a category member. For those unsure about their position in a field, status anxiety will push them to blend in and follow the crowd. For those who are secure in their identity and membership in a field, they will locate themselves within the field but also position themselves as distinct from the rest. Having established their credibility, they move to stand out. They are not captured by the herd but rather derive their advantage from not being beholden to standard wisdom. They understand what others value, which allows them to be regarded as legitimate, but they derive advantage from differentiation. For those who do not derive value from membership in a field, quite often those at the bottom of a status hierarchy, they are free to deviate from and not regard the rules of the community.

Such a process of differentiation presumably leaves its trace in the configuration of a field: the desire to become recognizable as a member of a social category drives the majority of field members to closely adhere to core principles of their particular domains with little motivation to adopt elements from other meaning systems or form ties across domain boundaries. Only those organizations at the low and high end of the status spectrum pursue distinctive discursive positions and form ties to other renegades. Their overall impact serves to partition fields into largely separate spheres with elites at the pinnacles of a hierarchy and those outside the status regime also free to pursue their own agendas.

Recombination can be a form of bricolage, whereby the practices or habits of others are incorporated into an organization's repertoire (Douglas, 1986). Such moves have been described by Burt (2005) as an import-export activity, and borrowing from others can lead to considerable profit. Such repurposing can, at times, result in unexpected gains, as familiar tools can be put to new purposes when in the hands of others (Padgett & Powell, 2012).

Powell and Sandholtz (2012) analyze this process of recombination with the example of the transposition of the academic laboratory from its origin in the university to the realm of commercial biotechnology firms. In the 1970s, strides in the understanding of molecular biology resonated beyond the academy, catching the attention of venture capitalists that saw profit potential in the new gene manipulation technologies. Amphibious scientist-entrepreneurs created the first generation of biotechnology companies using the scientific norms and practices they were familiar with from the university and combining these with modes of commercial enterprises. These science-based commercial entities had lasting repercussions in the original domain of their founders: universities

embraced patenting as an indicator of scientific output and in time celebrated academic entrepreneurship.

Biotechnology firms exemplify how connections between formerly separate communities facilitate the recombination of elements from a diverse origin. Cross-domain ties allow for the flow of practices, concepts, and ideas, enabling field members to adopt elements from diverse meaning systems and repurpose them for their objectives. This transposition and translation activity alters the discursive pattern of a field, which becomes diversified rather than being segregated by domains.

Integration occurs when broader trends influence organizations, prompting members of a field to use common practices and thus become more similar. A burgeoning line of research in the world polity tradition argues that the building blocks of modern organizations are now more available than ever and are ready for assembly by organizations of all stripes, from schools to religious sects and from NGOs to large corporations. Bromley and Meyer (2015) see cultural rationalization as the guiding force in this process of creating a world of organizations. To be sure, such organizing invites internal conflicts and even competing purposes; nevertheless, the common elements of modern organization are deeply integrated both cognitively and relationally. Some see meta-organizations, i.e., large, influential international associations, as driving this trend (Ahrne & Brunsson, 2008), whereas others emphasize the role model function of international nongovernmental organizations (Djelic & Sahlin, 2006).

Whereas fields shaped by differentiation and recombination processes exhibit a plurality of meaning systems, integration fosters the prominence of an overarching narrative. This narrative affects organizations irrespective of their original orientation and form, thereby creating a radial convergence on a common interpretation. Through such contacts, management trends, rather than being mere fads and fashions, can develop an allure that may affect an entire field. Some argue that the nonprofit sector's increasing professionalization and rationalization exemplifies such a development (Frumkin, 2005; Putnam, 2007).

The three social processes — differentiation, recombination, and integration — have varied effects on both the structure and discourse of organizational fields. Differences in the degree to which organizations are open to adopt elements of one or more meaning systems are reflected in their distribution across the field's cultural space, which may take various shapes ranging from a broad scattering to concentrated discourse clusters. The resulting variations in cultural distance between participants in turn affect the likelihood that relations may be forged, with those in close vicinity being more prone to connect than those located across the expanse of the topography of meaning. These cultural and relational structures, in turn, present different paths through which concepts and practices may travel, thus shaping processes of adaption, diffusion, and translation. Having outlined these dynamics conceptually, we turn to visualization and data simulation as means to explore how the theorized social processes manifest in different field configurations.

SIMULATING AND VISUALIZING MEANING SYSTEMS AND RELATIONAL STRUCTURE

We distinguish three aspects of organizations: form, social relations, and cultural expressions. Organizational form includes such attributes as legal status, age, size, workforce composition, and division of labor. Information on these features can be extracted from databases or manual classification of self-presentations (for instance, business reports or websites). The social relationships of an organization influence its position and role in an organizational field. Cultural expression reflects an organization's position in the discourse within an organizational field (Anheier, Gerhards, & Romo, 1995). Conceptually, organizational form, social relations, and cultural expressions can be studied independently of each other. Empirically, the variation of links among forms and relations, relations and culture, and culture and form creates different field configurations that we want to characterize.

Data on the relationships between organizations can be harder to obtain than data about form. A typical means to access connections, collaborations, or contracts between organizations is to ask survey questions, and code press releases, directories, or financial reports. Such sources often do not include new entrants or smaller organizations. New approaches to extract references, endorsements, or even interactions from social media data (for example, from e-mails, websites, and rating services) offer possibilities to collect information without researchers' or participants' interference with the data collection. Whether the relational data is manipulated further – for instance, by computing measures like co-occurrence or structural equivalence – or used in raw form, depends on particular research questions (see Salganik, 2017).

Cultural expressions can be found in press releases, reports, and websites that are issued by the respective organization (Meyer & Höllerer, 2010). Although these sources may change over time, symbols in artifacts such as logos may reflect an organization's long-term cultural position (Drori, Delmestri, & Oberg, 2016). After extracting and categorizing cultural expressions, an additional data manipulation step might be necessary to identify common themes or topics and an organization's adoption of these topics. Again, as with form and relations, it is crucial to keep organizations discernible as the unit of analysis for cultural expression.

By using organizations as the unit of analysis, all three data layers of form, network structure, and cultural expression can be studied jointly. A visual analysis can serve as a starting point for such a joint analysis, as it helps to create first hypotheses.

Mapping Relations and Meaning System

To map organizational form, discursive position, and network location, we developed a specific network visualization incorporating a topography of meaning (Korff et al., 2015). The starting point is a canvas on which the discursive

positions of organizations are mapped. For settings with three discourses, we use an equilateral triangle in which each pinnacle represents a theoretical position of an organization that only speaks in one discourse (100% of discourse). Each other point on the plane of the triangle reflects a discursive position where organizations combine expressions from different discourses ($x\%$ of discourse one, $y\%$ of discourse two, $z\%$ of discourse three, $x + y + z = 100\%$). The center of the triangle, for example, is constituted by an equal amount (33.3%) of three discourses. In the triangle, we position organizations according to their participation in discourses – for instance, by using specific keywords on their websites. The distribution of organizations on the topography uncovers certain uses of the meaning systems provided by the discourses. For instance, if many organizations lump together in one area, this reflects a shared meaning system that all constituent organizations ascribe to. If different domains of the topography are populated while the terrain in the middle is empty, we observe a field segregated into distinctive discourses.

In the next step, the organizations are drawn with different shapes (for instance, circles, triangles, or rectangles) depending on their form. Additionally, these forms were previously connected to a specific discourse – for instance, firms to a managerial discourse and universities to a scientific discourse – and colors or shapes can represent the heritage of organizations. The lightness of color can show whether a form has transitioned into a new area of the topography. By highlighting the movement of organizations from one discourse to another, processes of change where forms are at least partly redefined can be reflected through the visualizations. By looking at the distribution of shapes, one can observe whether organizations of a specific form remain in their original domain whereas others adopt a different discourse.

Finally, we draw the relationships between organizations and increase the size of shapes according to the extent of organizations' connections. Doing so allows us to compare connectivity across the topography of meaning, thus offering insights to the relational structure of a discursively patterned field. For instance, one can observe whether densely linked clusters are located in the same area or scattered across the whole topography. By utilizing standard network analytical concepts, including components, clusters, and bridges, we can analyze whether fields are segmented into separate enclaves with or without social connections bridging between discourses (Friedman, 2011).

Simulating Discursive Positions and Social Relationships

We simulate all three aspects – form, relations, and discursive position – for a given set of organizations. The simulation algorithm performs three steps:

Step 1 – Creation of organizations. In the first step, three groups of organizations representing different organizational forms are created. To keep the

simulation straightforward, we have a corresponding number of organizations for each form.

Step 2 – Discursive positions of organizations. For each group of organizations with the same form, we assign percentages that represent the extent to which this group is open to adopt concepts expressed by other organizations. For example, form A might be open to adopt 20% of the linguistic expressions of discourse B, 10% of discourse C, and maintain 70% of its own. For each organization in each group, we vary the group's position randomly so that each organization receives an individual discursive position influenced by the openness of its form. By applying a Gauss randomization, we assure that extreme deviations from the form's openness to adopt are less likely to occur than smaller deviations (Forbes, Evans, Hastings, & Peacock, 2010).

Step 3 – Assignment of relationships. In the last step, we assign a fixed number of bidirectional relations between pairs of organizations. To create these relations, we randomly select two organizations, measure the distance between their discursive positions, and multiply the distance with a random number (again based on a Gauss distribution with a mean of 1.0). If the resulting product is under a threshold for cultural distance above which relationships are unexpected, then a relation is established. This algorithm reflects homophily: the lower the distance between two organizations, the more likely they will form a social relationship (McPherson et al., 2001).

RELATIONAL AND CULTURAL FIELD CONFIGURATIONS

We ran thousands of simulations in which we varied the degree of openness to adopt one or two other discourses for each of the three forms. For the purity of the argument, we ran the simulations with a fixed level of homophily. We then visualized the results and selected interesting, prototypical field configurations.

Differentiation among Discourses

Simulations in which openness to adopt concepts from other discourses is low for organizations of all three forms produce maps characterized by distinct and segregated clusters of organizations (see Fig. 2, for example). In these configurations, organizations are hesitant to adopt concepts from other discourses, irrespective of their form. The result is that organizations pick positions in the area near the pure representation of the discourse of their respective heritage. As a consequence, the cultural space between the pure forms is nearly empty. Only a small number of outliers try to combine their original expressions with influences from one or two other discourses. Looking at the distances between

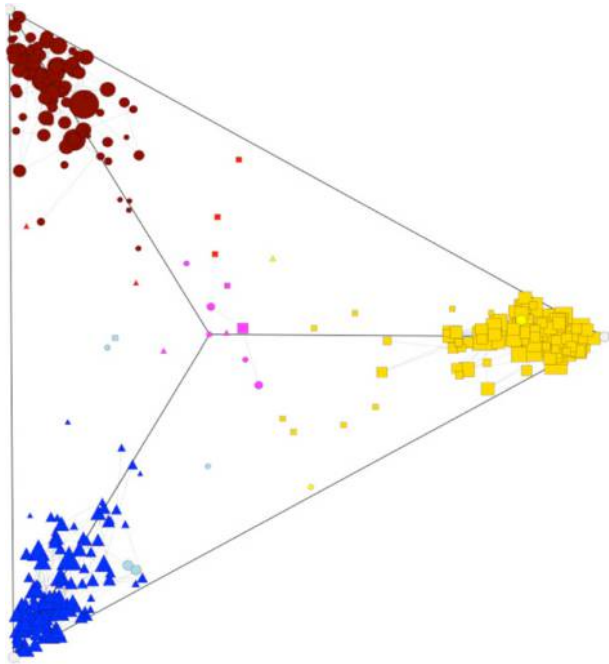


Fig. 2. Differentiation.

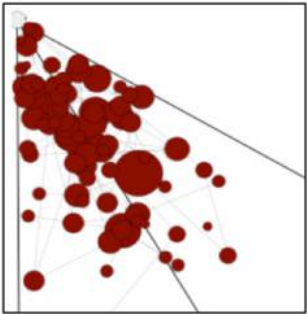


Fig. 3. Center-Periphery Structures within One Group (Zoom).

organizations of different forms, we observe that they are too vast to be bridged under the assumption of homophily. Even the few outliers are too distant from others to establish relationships.

When we magnify one of the three components in the pure discursive areas (see Fig. 3), we observe densely linked clusters of organizations of the same form. Although they are members of the same component, the number of

references to other peers varies. Despite the similarity in form, a type of status competition with a center-periphery structure is apparent.

In each of the three groups, form and relationships are strongly linked; none of the organizations have connections to organizations of different forms. The lack of openness to adopt concepts from others resulted in no organization significantly changing its cultural expression. Instead, the link between culture and form is strengthened. In sum, the simulation reveals a constellation with an internal status differentiation but with no interconnections.

The concentration into clusters with little bridging reflects a field shaped by differentiation. The simulated map highlights how a majority of members cohere together, creating distinctive and concentrated discourses from which few field members deviate. Those that do are solitary, isolated both from their peers in form as well as from other discursive outliers. This relational pattern suggests that deviation in this case is a function of low status and recognition. Prominent and highly connected field members do not seek exceptional positions or explore commonalities across domain boundaries but remain firmly embedded in the core discourses. With exploration left to those on the low end of the reputation scale, the chances for the emergence of a common discourse and domain-spanning network appear to be slim.

Recombination of Discourses

With a moderate increase in the degree to which organizations are open to new concepts, markedly different maps are generated. By adopting content derived from other discourses, organizations populate larger areas of the topography than in the previous simulations. Some field members may adopt a new discourse to the extent that their form receives a new meaning. The resulting hybrid form combines elements of different ways of organizing and bridges domains (Pache & Santos, 2013).

In Fig. 4, various hybrids – represented in a lighter hue – populate the area between the top and bottom right corner of the triangle. For the sake of illustration, we observe nonprofits that originally contribute to an associational discourse involved with managerial topics, whereas companies display an associational language. If two differing forms adopt one another's discourse, both forms move into a space sharing these discourses. In this area of overlap, we find organizations of different forms positioned next to one another. When organizations move closer discursively, their reduced cultural distance facilitates the formation of ties between them. This recombination of “old” and “new” discourses enables them to become brokers between discourses and forms. Lacking this cultural proximity, the organizations in the left bottom corner of the triangle are focused exclusively on their original discourse, thus remaining spatially separate as well as disconnected.

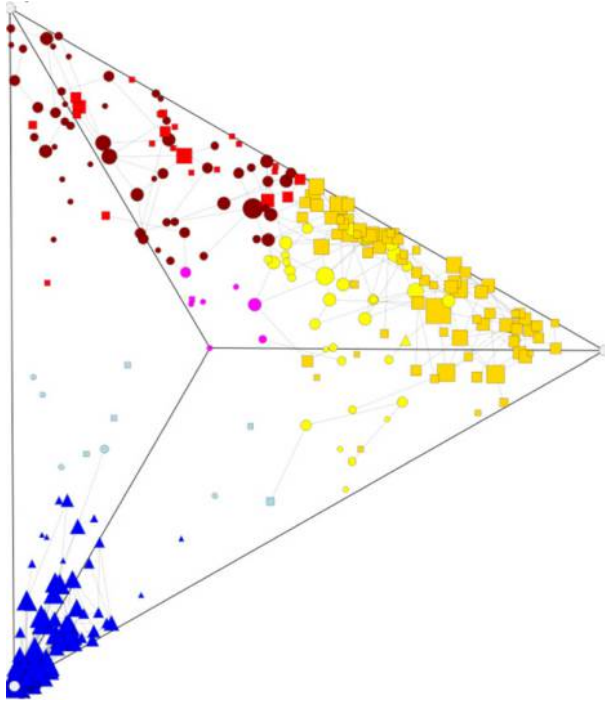


Fig. 4. Recombination.

As average openness to other discourses is increased across all organizations, irrespective of their form and original discourse, configurations emerge that resemble the empirical pattern observed for the issue field of social impact analysis. Fig. 5 represents an exemplary simulated map of a field in which the center of the triangle – the interstice between discourses – is populated by organizations that mix all three discourses with nearly equal amounts, while others remain true to their original discourses. With organizations distributed across the surface of the triangle, cultural distances between organizations are low. Almost all organizations have neighbors in their discursive vicinity and few unpopulated expanses remain on the topography. Given such short cultural distances, connections are easily forged. The simulated conditions of moderate openness across all organizational forms and discourses thus produce a scenario in which the field is spread over an array of discursive positions and is densely connected.

Of particular interest in this simulation is the set of organizations in the center of the triangle. Fig. 6 shows that organizations of all forms – represented by different shapes – appear in nearly equal numbers and are well connected among one another. Cultural proximity resulting from a mutual openness

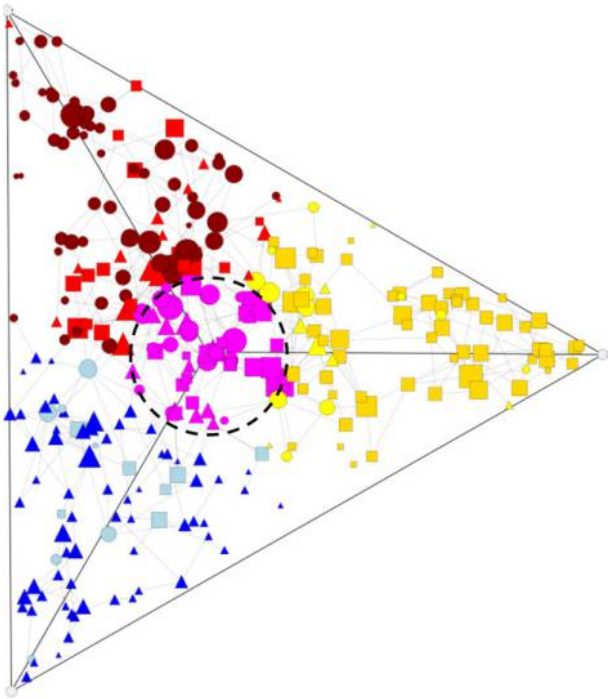


Fig. 5. Interstitial.

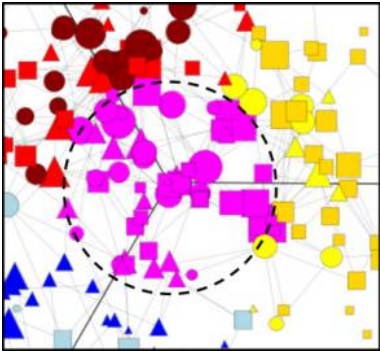


Fig. 6. Interstitial Organizations (Zoom).

toward their respective discourses allows the organizations in the interstice to speak in a common language and to form relationships, despite any differences in form. This inconsequentiality of organizational form distinguishes the interstitial scenario from the simulation of differentiation, as shown in Fig. 2 and

described above. Under conditions of low receptivity to other discourses, organizations remain in their original domains, primarily connected to organizations of similar form. In contrast to differentiation into separate homogenous clusters, we now observe that organizations have unique compositions of discursive neighbors of different forms with whom they are connected. Particularly in the interstitial zone, we find a heterogeneous group of organizations that speak in a common language and are densely interconnected, not unlike the interstitial community we observe empirically for the issue field of social impact analysis.

In correspondence with the empirical example, the simulated population of the interstice is not only internally cohesive but also well connected through multiple paths to organizations that remained positioned in the original domains. These external linkages connect organizations across the triangle and allow for content to traverse between discourses. The organizations in the interstitial zone serve as both translators that facilitate diverse contacts in their neighborhood and bridges for ideas and practices to flow from one discourse to another.

Developing One Integrated Discourse

In the prior simulations, organizations' receptivity to other discourses was mutual. Either all organizations were open to adopt new content or – as displayed in Fig. 4 – organizations of two discourses were equally oriented toward one another, while neglecting those that did not reach out. Eliminating such mutuality in the simulations creates different topographies. Particularly interesting are constellations in which two organizational forms have a high degree of openness to the concepts of a third form, while this third form's openness is low.

One example of such a map is pictured in Fig. 7, which shows that organizations open to adopt new concepts have moved toward the discursive domain characterized by low openness. Drawing organizations away from their original discourses, this one-sided attraction leaves areas of the topography vacant. Whereas the simulations discussed previously produced such vacancy at the center of the topography, namely under conditions of overall low openness and differentiation into separate clusters (Fig. 2), we now find areas associated with the original discourses left sparsely populated. The cultural content these discourses have to offer is not taken up and is even abandoned in favor of an alternative, more prominent discourse. This dominant discourse attracts organizations, irrespective of their form and origin, and leaves little leeway for neither resistance nor combination with other discourses.

As a result of such attraction, organizations of different origins have come together in the realm of the dominant discourse, where they form connections with one another (see Fig. 8). Through the increasing alignment of

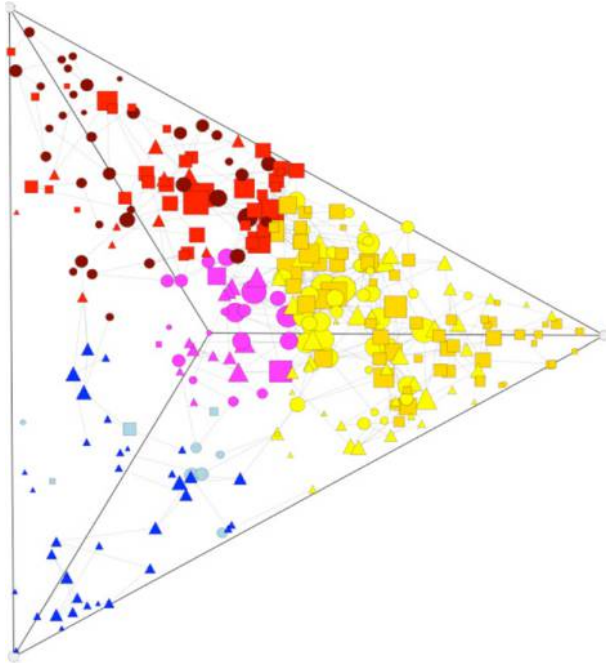


Fig. 7. Integration.

organizations in terms of their cultural expressions, the field as a whole becomes more diverse in form. In fact, dense connectivity, together with a shared culture, may function as a kernel for the emergence of a new organizational field when organizations of different forms begin to interact with one another and exhibit a shared meaning system (DiMaggio & Powell, 1983).

The degree of openness of the original three forms of organizations and whether attraction is mutual or one-sided influences where such integration manifests. If, as in the simulation described above, one discourse is taken up by all organizations irrespective of their form and origin, while other discourses are neglected, a cluster forms in the realm of this prominent discourse. Members of this cluster are able to converse with one another as they have attained fluency in the existing host language. Integration, in such a case, becomes synonymous with assimilation. A different scenario manifests if openness is mutual and not focused on any one discourse exclusively. Under such conditions, integration occurs in the center of the triangle, on equal terms for all forms. Together they create a language that is a composite of the original discourses and, as such, new to everyone. In this setting, the interstice is once more the area of concourse, not just for a few pioneering organizations that

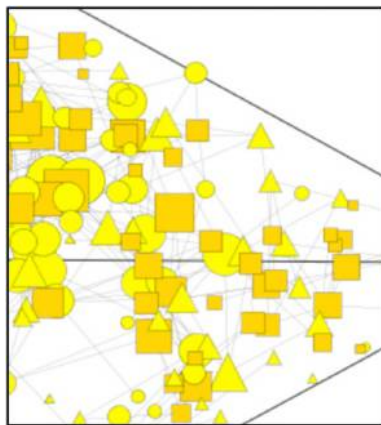


Fig. 8. Integration of Forms (Zoom).

form an interstitial community as discussed above but also for the majority of field members. Whether an interstitial community may be a prelude to integration and possibly a field formation on neutral common ground depends on the pull of the new composite discourse.

When comparing the three maps that exemplify differentiation (Fig. 2), bridging in the interstitial zone (Fig. 5), and integration (Fig. 7), we make the following observations: the coverage of the topography is highest when openness is moderate for all three groups as in the interstitial simulations. When openness is low, organizations end up in communities in their respective discourse; when it is high, organizations come together in one area of the topography. The wide dispersion of organizations in the interstitial scenario reduces the average distance between organizations and makes it easy to create social relationships. As a result, the network structure in this setting differs drastically from the other two: While the first and last simulations show center-periphery structures in components around shared discourses, the interstitial scenario contains long paths connecting organizations of different discourses. Finally, we observe in the last two scenarios many cases in which organizations moved from their heritage into another discourse and established multiplex ties to organizations of different forms.

The fact that we can observe differences between the interstitial and the integrated constellations is a benefit of combining cultural and relational analysis on the level of fields. To exemplify this point, we select two dense areas from both settings. Drawing these areas with a pure form and network approach (first row in Fig. 9), we observe that both segments entail well-connected organizations of different forms. Their level of connectivity varies, but the variation is independent of their organizational form. When we highlight different discourses (second row in Fig. 9), it becomes obvious that the mixed areas are

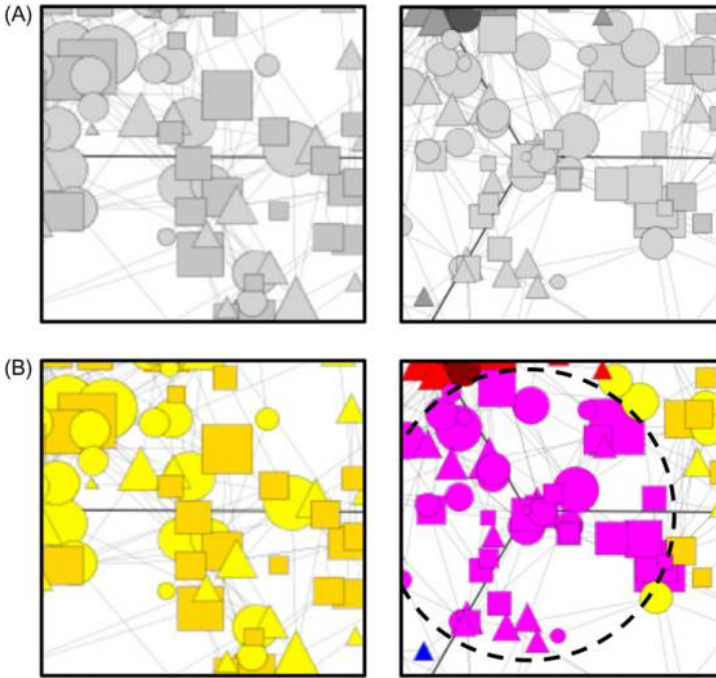


Fig. 9. Dense Areas in the Integrated (Left) and Interstitial (Right) Constellation. (A) Network and Form. (B) Network, Form, and Cultural Expression.

located in different regions of the topography. In the integrated example, the discourse of one form dominates the area, while all organizations adopted new discourses and bridge between these discourses in the interstitial setting. To conclude, the same network constellation can have different meanings depending on its position in the cultural space.

The mapping of form, heritage, and relationships enables us to recognize differences among individual organizations. We selected, for example, one dense area of the integration scenario (Fig. 10D). When we reduce this area to the cultural positions (Fig. 10A), we observe a more or less even distribution of organizations in the cultural space. When adding information on organizational forms (Fig. 10B), we see the heterogeneity of forms in the same cultural space. The lighter color hue makes those organizations that adopted a new discourse stand out. Instead of adding form to the discursive position, we combined position and network information (Fig. 10C). Now differences in the connectivity and status of organizations become visible. By combining cultural expressions,

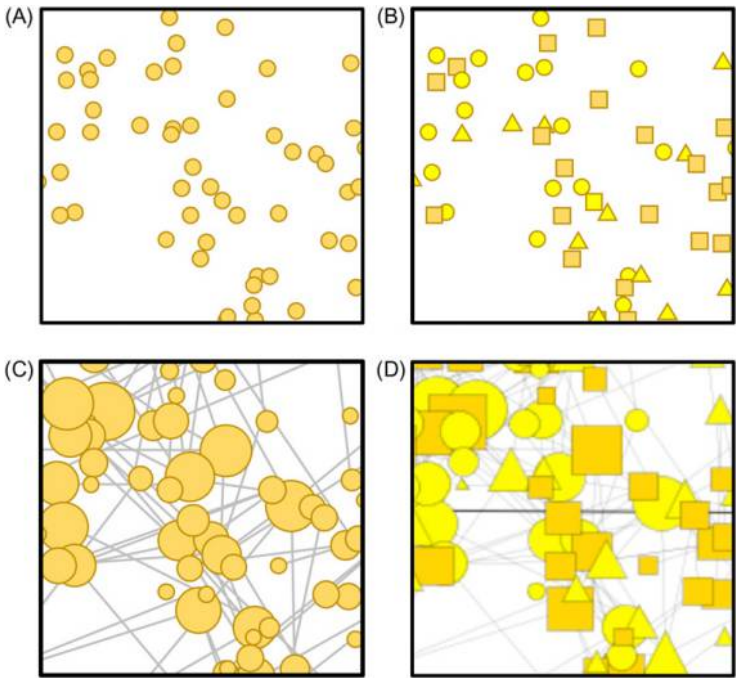


Fig. 10. Combining Levels of Analysis. (A) Cultural Positions. (B) Cultural Positions and Form. (C) Cultural Positions and Network. (D) Cultural Positions, Network, and Form.

forms, and relations (Fig. 10D), we get the full picture of a densely connected area of different forms that share expressions.

To conclude, by combining the relational structure and meaning system in this kind of mapping, interdependencies and frictions between organizational forms, neighboring connections of organizations, shared expressions, and bridging paths between discourses become visible.

DISCUSSION AND IMPLICATIONS

Our approach to combine meaning systems and relational structures consists of three steps: First, we simulate social mechanisms within fields by changing the positions of organizations on a discursive topography and by assigning social relations. Second, by using the organization’s discursive position as location for nodes connected via relationships similar to typical network visualizations. Third, these visuals help us to understand the resulting configurations of

organizations and to compare the mechanism with each other. Combined, these three steps – simulation, visualization, and interpretation – contribute to the study of field configuration mechanisms, exemplify how interpretative approaches can be applied to network analysis, and provide a new approach for diffusion research.

Study of Field Configuration Mechanisms

To study field configuration mechanisms, we reduced the theoretically described mechanisms to two axiomatic assumptions about the openness to adopt other discourses and the connection between similarity and connectivity. We went further and did not vary the degree of homophily, which assumes that a rise in similarity between organizations increases the likelihood of social relationships. Although this is a sharp reduction of complex social mechanisms to only two assumptions, the simulations nonetheless resulted in highly diverse field configurations that match our theoretical expectations. Different degrees of openness for adoption produced configurations of differentiation, recombination, and integration. If all forms of organizations show a low level of openness, we observe: differentiation; under varying conditions of moderate openness, we see different forms of combinations or hybridizations of discourses and sometimes the emergence of an interstitial community that bridges across domains; finally, in case of two forms with a high openness and one with a low degree, the result is integration in the discursive domain of the unwilling organizations. As in other simulations (see [Pancs & Vriend, 2007](#); [Schelling, 1969](#)), the reduction of complex social mechanisms to simple parameters turns out to be a strength.

That the same simulation algorithm produces different outcomes for varying parameters allows us to generate many configurations that can be systematically compared. The second step of visualization makes the results tangible and legible even without further statistical analyses. When comparing configurations, we observe changes in the population of the topography. We find spaces settled under conditions of differentiation that are only weakly populated in an integration scenario. Such shifts in the population of the topography of meaning suggest there is a corresponding broadening or narrowing of debate (for instance, [Meyer & Höllerer, 2010](#)). At the same time, we can compare differences in the density of relationships between snapshots: If a densely connected area becomes disconnected, it is a clear sign for an increase in competition and a decrease in mutualism. Together, discursive shifts and relational changes help us understand the dynamics in a field even beyond the basic mechanisms of differentiation, recombination, and integration.

The simulated maps serve, furthermore, as a reference point for interpreting our real-life data. When comparing the visualization of our empirical case (see [Fig. 1](#)) with simulated cases ([Figs. 2, 4, 5, and 7](#)), three observations stand

out: One, the highest overlap between our real-life data and the simulated maps is in the interstitial example. In real life and in simulation, we observe three communities in the three corners of the triangle and one interstitial community in the center. The high similarity leads to the conclusion that many organizations in our empirical setting seem to have a medium degree of openness to adopt new concepts. Second, differences can be observed too. In our empirical example, we observe some organizations that bridge between the associational and the managerial discourse without referencing the scientific discourse and others that connect the associational and the scientific discourse while omitting managerial terms, but there are no direct bridges between the managerial and the scientific discourse. In our simulations, we do not observe such a specific mode of combination. Instead, we find bridges between all three discourses. The lack of direct relationships in the empirical case could be induced by a higher degree of competition between managerial and scientific organizations. A third observation is that the density of the interstitial zone in the center of our empirical example is higher than in the simulations. This higher-than-expected density could be a sign of structuration processes that go beyond the simulation processes. As the combination of the three discourses is a relatively new phenomenon, the organizations that pick these positions might be more willing to cooperate and thus express a higher level of homophily in the interstitial zone.

The observed deviations between simulation and real data might be interpreted as simulation errors. Instead, we think it is helpful to see that the similarity between our simulations and the empirical data is high, even when the underlying simulation model is relatively simple. Thus, we can distinguish between general aspects that can be deducted from a model and specific aspects that might be linked to variations in competition and homophily in different zones of the topography.

Meaning of Relationships, Types of Organizations, and Individual Organizations

Besides the analytical possibilities of simulation and visualization, the mapping of networks on a topography of meaning illuminates the discursive embeddedness of core elements of social networks.

In many network analyses, the meaning of a social relationship is stable (a friendship is a friendship, a collaboration is a collaboration, etc.). By layering relationships of the same type above different discourses, it becomes obvious that there might be variation. For instance, a cooperation among nonprofit organizations set up to increase the impact on social justice might be handled differently than a cooperation between companies that develop a new tool for impact measurement. Although the first

cooperation might be highly influenced by shared values and expectations, the second cooperation might be coordinated as an economic exchange. Even when the collected data does not contain such variations in its meaning of relationships, by layering the data on top of the topography researchers are forced to question the similarity of meaning of relationships in different areas of the topography.

When we look at the distribution of organizations of the same form, we already recognize that some forms spread into many areas, while others stay in one discourse. The breadth of the area covered by organizations of the same form questions, again, the assumption that organizational forms have a stable meaning. Instead, with the openness to adopt concepts of other discourses, we observe hybrid forms mingling between established and new elements. In our empirical example, we observe consultancies, which adopted the language of nonprofits extensively resulting in discursive positions atypical for other consultancies. Similarly, commercial social enterprises look different from traditional businesses and nonprofits. As for relationships, the location of forms on the map can help raise questions about the meaning of a form.

Finally, the representation of relationships and forms as embedded in a discursive space renders the individual local network of each organization more informative than in a typical network visualization. For instance, if an organization shares two relationships of the same type with other organizations, these relationships can differ extensively if one tie connects with an organization in close discursive proximity while the other spans a larger distance on the topography. Direct neighbors share a common language and will easily understand one another, whereas distant field members might need additional explanations to interpret each other's communication. By taking these discursive distances into account, the maps capture the context in which sense making processes unfold. Specifically, the visualizations offer a structural approach that situates activities of meaning construction by organizations and their members into the broader discursive and relational structure of the field. This focus on the relational and discursive context of local sense making reflects earlier ideas about analyzing intersubjective meaning generation: Weber, in particular, saw social relationships as central units of analysis to be combined with cultural and social explanations (Ringer, 2009; Weber, 1978). Our method expands this approach to large datasets of unstructured texts and the references among organizations included therein. The World Wide Web, corporate intranets, large social media platforms, and similar digital networks provide information on both cultural and relational content. From such material discursive positions of organizations and the relationships among them can be extracted and used as input for our visualization. The mapping in combination with a structural analysis then helps to identify complex patterns in large datasets (Franzosi, 2004).

Diffusion of Concepts and Practices

In the previous sections, we discussed how the different simulation maps and/or real-life maps enable us to observe relational and discursive configurations and individual organizational contexts. We now turn to the question of how change is introduced into this picture: How can we study the diffusion of concepts and practices with this approach? In diffusion studies (for instance, [Centola, 2015](#)), social relationships are often seen as the conduits for the spread of ideas and knowledge ([Haythornthwaite, 1996](#)). From this perspective, social relationships are like railroad tracks, organizations have fixed positions like cities, and communication between organizations functions like trains that transport ideas as cargo from city to city. Some aspects of this underlying transportation metaphor have been criticized before. In contrast to physical goods, concepts are interpreted by the receiver based on his or her position and knowledge. Through interpretation the receiver influences the transmitted message ([Luhmann, 1992](#)). The communicated content gets edited and changed even further when it is not directly communicated from one organization to another organization, when other organizations are acting as mediators ([Czarniawska & Sevón, 2005](#)). These mediators translating the message add changes, whether they want it or not ([Czarniawska & Joerges, 1996](#)). Most quantitative diffusion studies have problems addressing this critique ([Meyer & Strang, 1993](#)). Simultaneously, those researchers who pointed out that editing and translation studies should be studied have not offered solutions to measure translation and editing processes ([Sahlin-Andersson, 1996](#)).

In the previous paragraph, we explained that the intertwined mapping of meaning structures and relational structures is helpful in interpreting the context of relationships, forms, and organizations. The same argument applies to the diffusion – or traveling – of concepts. The meaning of a concept is (at least partly) defined by its surrounding discourse. For instance, if the concept of social impact is embedded in a discourse about charity, social justice, and societal values, social impact is a soft concept that is hard to quantify. When social impact is framed with data collection, surveys, and analysis methods, the same terms gain a different meaning and become (slightly) different concepts. If a concept diffuses across a larger field spread above a discursive topography, its diffusion is not a simple transportation. Instead, either the diffusing concept changes its meaning partly to reach organizations in distant areas or the adopting organizations change their positions to pick up a concept which they did not use before.

The mapping approach helps to quantify the necessary adaption of a concept. The Euclidian distance between two organizations – A and B – reflects the discursive or cultural distance. This is important because typical networks reflect only the social distance (path length) by counting the number of relationships needed to transmit an idea or concept between two organizations. They

ignore the cultural distance. Another difference is the handling of isolates: If an organization has no relationships, its position is undefined in typical network analysis because a transmission of information alongside social relationships is not possible (Laumann, Marsden, & Prensky, 1989). In our approach, such relationally isolated organizations have clear positions based on their discursive behavior. This small adjustment to standard network analysis makes a substantial difference when we analyze weakly structured emerging fields where some actors have established positions but are ignored by their potential peers, or when we want to study the diffusion of concepts. We can include these isolates that pick up concepts in the media and become part of the field, although they are not yet connected to others.

The discursive distance influences the amount of necessary change in meaning when organization B adapts a concept that A uses. In other words, this distance reflects the degree of necessary translation and editing (Sahlin-Andersson, 1996). Whether this distance can be bridged via social relationships is visible in the map too. Either there is a direct relationship between A and B that allows the concept to travel or there are paths crossing other organizations to connect A and B. These paths might not take the shortest route, often the sum of the discursive distance of a path might be much higher than the direct discursive distance itself. If a concept travels such a winding path, it is likely to be changed even further.

CONCLUSION

We reject the idea that discourse and relationships are different realms; they are, in fact, co-constitutive: first, cultural cues condition the relations organizations forge, the invitations they accept, and the relations that endure (McLean, 2007); second, social affiliations are not only conduits for flow of concepts and ideas but also networks of meaning (Breiger & Mohr, 2004; Mische, 2003; Pachucki & Breiger, 2010). Combining both processes, one could formulate for organizations in a field: whereas actors make relations in the short run, in the long run, those relations constitute actors (Padgett & Powell, 2012, p. 2), and for the field level one could say that the meaning of foundational elements of social networks – types of social relationships, organizational forms, and concepts flowing – is shaped by the discourses that these networks connect.

Our approach offers one solution to map social relationships of organizational fields (Zietsma et al., 2017) or movements (Diani, 2015) on meaning systems. In doing so, the cultural embeddedness of relationships, organizations, and content becomes recognizable in visual analyses of networks. At the same time, large structures of fields and organizations' individual relational and discursive positions become visible. By adding a layer of meaning to social networks, the resulting topography opens new perspectives for institutional

analyses of organizational fields. Discursive positions and the relationships within a field can be studied systematically to identify mechanisms such as differentiation, recombination, and integration in organizational fields. In turn, the simultaneous attention to different discourses and connecting paths within fields holds the potential to revive research on the diffusion of concepts (Meyer & Strang, 1993), in that it allows quantifying the changing meaning of travelling concepts (Czarniawska & Sevón, 2005).

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